Linear Algebra and Matrix Analysis

Weiguang Liu

Collections from Linear Algebra Done Right, Strange's Lectures.

Horn and Rogers' Matrix Analysis, the first three chapters are standard materials. Then there are theories on

1. Hermitian Matrices

2. Matrix Norm Pinite and matrices with positive elements.

Then we have Matrix Calculus book. We may delve into Random Matrix Theory later.

Contents

0.1 Finite-Dimensional Vector Space and Inner Product Spaces 2
0.2 Invariant Subspaces and Spectral Theorems 2
0.3 Canonical Forms 3
0.4 Hermitian Matrix 3
0.4.1 Characterisation 3
0.4.2 Variational Characterisation 3
0.4.3 Eigenvalue Inequality for Hermitian Matrices 4
0.4.4 Complex Symmetric Matrices 4
0.5 Norms of Matrices 4

0.1 Finite-Dimensional Vector Space and Inner Product Spaces

0.2 Invariant Subspaces and Spectral Theorems

Spectrum. $\sigma(A)$ the set of all eigenvalues of A.

0.1 Proposition (About Matrix Transformation). *For $A = \bar{A}$ that's the complex conjugate of A, $Ax = \bar{A}\bar{x}$.*

For a polynomial $p(A)$, if $\lambda \in \sigma(A)$, then $p(\lambda) \in \sigma(p(A))$. If $\mu \in \sigma(p(A))$, then there exists $\lambda \in \sigma(A)$ such that $p(\lambda) = \mu$.

0.2 Proposition (Normal Matrix). *If $AA^* = A^*A$, then A and A^* share the same eigenvectors, the eigenvalues are λ and $\bar{\lambda}$.*

Proof. If $Ax = \lambda x$, let $B = A - \lambda I$, $x \in N(B)$. $\langle B^* x | B^* x \rangle = \langle x | BB^* x \rangle = \overline{\langle B^* B x | x \rangle} = \overline{\langle Bx | Bx \rangle} = 0$. □

0.3 Proposition. *Linear Operator on Inf Dim Vector Space might not have eigenvalue. Consider the right-shift operator $S(x_1, \ldots) = S(0, x_1, x_2, \ldots)$.*

0.3 Canonical Forms

0.4 Hermitian Matrix

0.4.1 Characterisation

For complex matrix $A \in \mathbb{M}(\mathbb{C})$, Hermitian matrices are like "real numbers to complex numbers" due to the following decomposition:

0.4 Toeplitz Decomposition. *For any $A \in \mathbb{M}(\mathbb{C})$, there exists unique Hermitian H, K such that $A = H + iK$. It can also be uniquely decomposed as $A = H + S$ where S is skew-Hermitian: $S = -S^*$.* ◇

0.5 Simultaneous Diagonalizable for Hermitian Matrix. *For a commuting class of Hermitian matrix, there exists one unitary U that simultaneously diagonalize $A \in \mathcal{F}$.* ◇

Proof. First, simultaneously make matrices upper-triangular. Then apply the spectral theorem to normal matrices. The Hermitian case is a special case. □

Then it's connected with Positive Definite matrices.

0.4.2 Variational Characterisation

0.6 Courant-Fischer. *Let A be Hermitian and $\lambda_1 \leq \cdots \leq \lambda_n$ be eigenvalues, for $1 \leq k \leq n$, and S a subspace of $\mathbb{C}^n$.*

$$\lambda_k = \min_{S : \dim S = k} \max_{x \in S \setminus \{0\}} \frac{x^* A x}{x^* x} \quad \textit{and} \quad \lambda_k = \max_{S : \dim S = n-k+1} \min_{x \in S \setminus \{0\}} \frac{x^* A x}{x^* x}$$

◇

Proof. Uses Rayleigh's Characterisation and subspace intersection lemma. □

0.4.3 Eigenvalue Inequality for Hermitian Matrices

0.7 Weyl's Theorem. ◊

0.8 Cuchy's Interlacing Theorem; Separation Theorem. ◊

0.4.4 Complex Symmetric Matrices

0.5 Norms of Matrices